Given: Due:	Further Maths (Core Pure) HW 02	Name:

Chapter 1 & 2: Complex Numbers: modulus-argument

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)

Useful resources: *Useful resources*: Your notes from recent lessons & your text book

Although most Complex arithmetic can be done on your calculator, show all steps to reach the answer & use calculator for checking results only.

Questions - do these on lined paper

1)

$$z_1 = 3\text{i} \text{ and } z_2 = \frac{6}{1+\text{i}\sqrt{3}}$$

(a) Express z_2 in the form a + ib, where a and b are real numbers.

(2)

(b) Find the modulus and the argument of z_2, giving the argument in radians in terms of π.

(4)

(c) Show the three points representing z_1, z_2 and ($z_1 + z_2$) respectively, on a single Argand diagram.

(2)

2) (i) The complex number w is given by

$$w = \frac{p-4\text{i}}{2-3\text{i}}$$

where p is a real constant.

(a) Express w in the form a + bi, where a and b are real constants. Give your answer in its simplest form in terms of p.

(3)

Given that arg $w = \frac{\pi}{4}$

(b) find the value of p.

(2)

(ii) The complex number z is given by

$$z = (1 - \lambda\text{i})(4 + 3\text{i})$$

where λ is a real constant.

Given that

$$|z| = 45$$

find the possible values of λ

Give your answers as exact values in their simplest form.

(3)

3) The complex number w is given by

$$w = 10 - 5\text{i}$$

(a) Find $|w|$.

(1)

(b) Find arg w, giving your answer in radians to 2 decimal places.

(2)

The complex numbers z and w satisfy the equation

$$(2 + \text{i})(z + 3\text{i}) = w$$

(c) Use algebra to find z, giving your answer in the form $a + b\text{i}$, where a and b are real numbers.

(4)

Given that

$$\arg(\lambda + 9\text{i} + w) = \frac{\pi}{4}$$

where λ is a real constant,

(d) find the value of λ.

(2)

4)

$$z_1 = 2 + 3\text{i}, \quad z_2 = 3 + 2\text{i}, \quad z_3 = a + b\text{i}, \quad a, b \quad \frac{z_1 z_3}{z_2}$$

(a) Find the exact value of $|z_1 + z_2|$.

(2)

Given that $w =$,

(b) find w in terms of a and b, giving your answer in the form $x + \text{i}y$, x, y $\frac{z_1 z_3}{z_2}$

(4)

Given also that $w = \frac{17}{13} - \frac{7}{13}\text{i}$,

(c) find the value of a and the value of b,

(3)

(d) find arg w, giving your answer in radians to 3 decimal places.

(2)

END OF HOMEWORK

Total = 36 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.